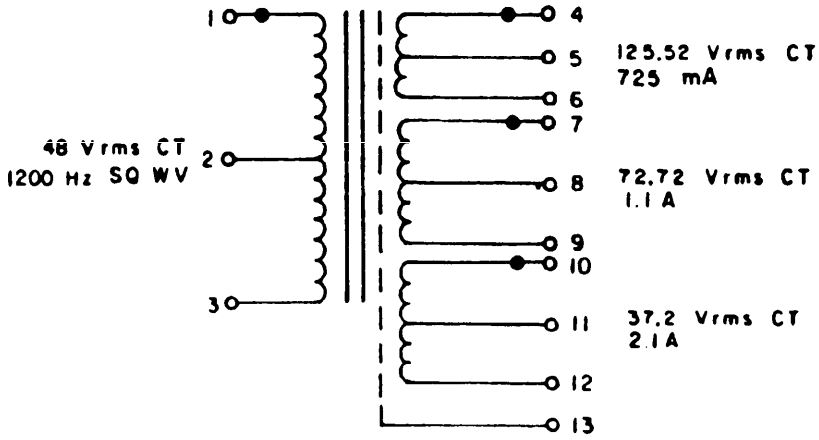


ALTITUDE: 100 k FT MAX
WORKING VOLTAGE (PEAK)
(1-3): 58 V
(4-5) & (5-6): 70 V
(7-8) & (8-9): 40 V
(10-11) & (11-12): 21 V

CIRCUIT DIAGRAM AND MARKING

INCHES	MM	INCHES	MM
.010	0.25	.29	7.4
.020	0.51	.30	7.6
.08	2.0	.40	10.2
.090	2.29	.75	19.0
.140	3.56	1.15	29.2
.144	3.66	1.188	30.18
.15	3.8	1.56	39.6
.21	5.3	1.790	45.47
.22	5.6	1.940	49.28
.25	6.4	3.125	79.38

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the top of the case.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS: (When numbers in parantheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage (1-3): 48 V rms ct, 1,?00 Hz square wave.

Volt-ampere rating: 190 VA.

Secondary voltage:

(4-6): 125.52 V rms ct.

(7-9): 72.72 V rms ct.

(10-12): 37.2 V rms ct.

Secondary current:

(4-6): 0.725 A dc.

(7-9): 1.1 A dc.

(10-12): 2.1 A dc.

Working voltage (peak):

(1-3): 58 V.

(4-5) and (5-6): 70 V.

(7-8) and (8-9): 40 V.

(10-11) and (11-12): 21 V.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Material: Epoxy.

Terminals: Solder terminal.

© (1, 3, 4, 5, 6, 7, 9, 10, 11, and 12): USECO (FSCM 15849) number 1281B or equivalent.

© (2, 8): USECO number 1282B or equivalent.

(13): Hadley (FSCM 72964) D lug or equivalent.

Height:

(1, 3, 4, 5, 6, 7, 9, 10, 11, and 12): 0.219±0.010 inch.

© (2, 8): 0.281±0.010 inch.

(13): 0.25±0.10 inch.

Weight: 0.55 kilogram maximum.

Altitude: 100,000 feet maximum.

Operating temperature range: -55°C to +170°C.

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Terminal strength: MIL-STD-202, method 211, test condition A, 5 pounds.

Dielectric withstanding voltage:

At sea level: 500 V rms.

At reduced barometric pressure: 300 V rms.

Electrical characteristics:

No load:

Condition 1: With 56 V rms, 1,200 Hz sine wave across (1-3):

Current in (1-3): 0.5 A maximum.

© Power in (1-3): 6.0 W maximum.

Condition 2: With 24 V rms, 1,200 Hz sine wave across (1-2).

Voltage across (4-6): 126.7 V ct ±1 percent.

Voltage across (7-9): 73.6 V ct ±1 percent.

Voltage across (10-12): 37.7 V ct ±1 percent.

Rated load: With 12 V rms ct ±0.2 percent, 1,200±40 Hz sine wave across (1-2).

Voltage ratio $\frac{V(4-6)}{V(1-2)}$: 5.23±1 percent

Voltage ratio $\frac{V(7-9)}{V(1-2)}$: 3.03±1 percent

Voltage ratio $\frac{V(10-12)}{V(1-2)}$: 1.55±1 percent

Electrostatic shielding: Voltage ratio of 5 to 1 at 20 kHz sine wave.

DC resistance:

(1-2) = (2-3): 0.06 ohm maximum.

(4-5) = (5-6): 1.7 ohms maximum.

(7-8) = (8-9): 1.0 ohm maximum.

(10-11) = (11-12): 0.28 ohm maximum.

Polarity: Additive, with terminals (3-4), (6-7), and (9-10) connected.

Series leakage inductance: 20 µH maximum at 0.1 V rms at 10 kHz measured across terminals (1-2) or (2-3) with terminals (4, 5, 6, 7, 8, 9, 10, 11, and 12) short circuited and terminals 1 or 3 as an open circuit. Series leakage inductance shall not exceed 20 percent of the value measured or 2.0 µH whichever is greater, when measured at terminals (1-2) or (2-3) with terminals (4, 5, 6, 7, 8, 9, 10, 11, and 12) and 1 or 3 as an open circuit.

Center-tap voltage unbalance: 24 V rms, 1,200 Hz sine wave across (1-2).

Condition 1: No load on secondaries.

$\frac{V(1-2)}{V(2-3)}$: 1.0±0.5 percent

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Condition 2: Only the one secondary being tested shall be loaded with rated load current with the other two secondaries open circuited.

$$\frac{V(4\text{-}5)}{V(5\text{-}6)} : \frac{V(7\text{-}8)}{V(8\text{-}9)} : \frac{V(10\text{-}11)}{V(11\text{-}12)} = 1.0 \pm 0.5 \text{ percent.}$$

Regulation: 5 percent, maximum from no load to full load.

Temperature rise: 45°C with 24 V rms, 1,200 Hz sine wave across (1-2) at an ambient temperature of 125°C. All secondaries shall be loaded simultaneously with their rated load. The transformer shall be mounted on a 3.5 x 10 x 1/8 inches thick aluminum plate.

Vibration, high frequency: MIL-STD-202, method 204.

Marking location: See figure 1.

Part number: M27/199-01.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - MC, AS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5950-0641-5)

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